

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 717 Algorithmic Game Theory — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Nisan, Roughgarden, Tardos & Vazirani — Algorithmic Game Theory (free PDF from the authors)	The primary text, and it is free. Modules 02 through 11 follow its organisation, and the chapters are by the people who proved the results.
Roughgarden — Twenty Lectures on Algorithmic Game Theory, with the free video lectures and notes	The best entry point, free in full. Modules 01, 04, 06, and 07 especially — and the lectures are worth watching rather than only reading.
Shoham & Leyton-Brown — Multiagent Systems (free PDF)	Modules 02, 10, and 12, free. Broader than the algorithmic texts, and the social choice chapters are excellent.
Krishna — Auction Theory	Module 07. The economics treatment, where revenue equivalence is derived properly. Library copy.
Roth — Who Gets What and Why	Module 09. Matching markets as actually deployed — kidney exchange, school choice, medical residencies — by the person who built several of them.
Brandt, Conitzer, Endriss, Lang & Procaccia — Handbook of Computational Social Choice (free PDF)	Modules 10 and 11, free. The impossibility results and the fair division literature, rigorously.

Module-by-module map

Module 01 — When the Input Has Interests

Resource	What it covers for this module
Roughgarden — Twenty Lectures, lecture 1 (free)	§§1 and 3 — the framing, with the sponsored search example developed as the motivating case.
Nisan et al., chapter 1 (free PDF)	§3's two questions , and the book's organisation follows that division.
Roughgarden — Algorithmic Game Theory, CACM (free)	§§1 and 2 — a short argument for why computer scientists need this, aimed at exactly this course's audience.
Varian — Position auctions (free preprint)	§2's first row — the ad auction analysed, which is the deployment that made this field commercially serious.

Module 02 — Games and Equilibria

Resource	What it covers for this module
Shoham & Leyton-Brown, chapters 3 and 4 (free PDF)	§§1 to 3 — the solution concepts, carefully distinguished, which is what this module is about.
Nash — Equilibrium points in n-person games (free)	§3 — one page, and the proof is the fixed-point argument in its original form.
Osborne & Rubinstein — A Course in Game Theory (free from the authors, after a one-step registration)	§§2 and 4 — rigorous, free, and unusually careful about what each concept assumes.
The Gambit game solver (free)	§4 — compute equilibria of small games, and watch it become infeasible, which is Module 03's point made experimentally.

Module 03 — Computing Equilibria

Resource	What it covers for this module
Daskalakis, Goldberg & Papadimitriou — The complexity of computing a Nash equilibrium (free)	§2 — the hardness result, with a simplified proof by the authors.
Papadimitriou — On the complexity of the parity argument (free)	§2 — where PPAD was defined, and the motivation is stated plainly.
Nisan et al., chapter 2 (free PDF)	§§1 to 3 — the computational treatment of equilibrium, including the algorithms that do work.
Roughgarden — lectures on complexity of equilibria (free)	§§3 and 4 — including the argument for why hardness matters for the concept's use.

Module 04 — The Price of Anarchy

Resource	What it covers for this module
Koutsoupias & Papadimitriou — Worst-case equilibria (free)	§1 — where the measure was introduced, in a load balancing setting.
Roughgarden & Tardos — How bad is selfish routing? (free)	§2's routing bound — with the latency function condition made explicit, as it should be.
Roughgarden — Intrinsic robustness of the price of anarchy (free)	§3's smoothness framework — and the extension to no-regret outcomes is the result that matters.
Nisan et al., chapters 17 to 19 (free PDF)	§§2 and 4 — the bounds for routing, load balancing, and network design.

Module 05 — Routing and Congestion

Resource	What it covers for this module
Rosenthal — A class of games possessing pure-strategy Nash equilibria	§2's potential function , in the original — three pages.
Braess (1968), translated (free)	§3 — the paradox in its original form, with the translation and commentary.
Roughgarden — Selfish Routing and the Price of Anarchy	§§1 to 4 — the book-length treatment, including tolls and Stackelberg routing. Library copy.
Nisan et al., chapters 18 and 19 (free PDF)	§§2 and 4 — potential games and the interventions, with the proofs.

Module 06 — Mechanism Design

Resource	What it covers for this module
Nisan et al., chapters 9 and 11 (free PDF)	§§1 to 3 — mechanism design and VCG, with the computational constraints treated seriously.
Vickrey, Clarke, and Groves — the three original papers	§3 — and it is worth seeing how differently the three authors arrived at the same construction.
Myerson & Satterthwaite — Efficient mechanisms for bilateral trading	§4 — the impossibility, in the original.
Roughgarden — Twenty Lectures, lectures 2 to 7 (free)	The whole module — and the development of VCG from single-item auctions is the clearest route to it.

Module 07 — Single-Item Auctions

Resource	What it covers for this module
Vickrey — Counterspeculation, auctions, and competitive sealed tenders	§1 — the second-price auction in the original, and the argument is the one in the callout.
Myerson — Optimal auction design	§3 — virtual values and the revenue-optimal auction. Library copy.

Resource	What it covers for this module
Krishna — Auction Theory	§§2 and 4 — revenue equivalence derived properly, with every condition made explicit. Library copy.
Kagel & Levin — the experimental auction literature	§4 — what bidders actually do, measured, including the winner's curse in the field. Library copy.

Module 08 — Combinatorial Auctions

Resource	What it covers for this module
Cramton, Shoham & Steinberg — Combinatorial Auctions	The whole module — the standard collection, with chapters on each of §3's routes.
Lehmann, O'Callaghan & Shoham — Truth revelation in approximately efficient combinatorial auctions (free)	§§2 and 3 — the single-minded case, and the monotonicity technique that makes approximation truthful.
Nisan et al., chapters 11 and 12 (free PDF)	§§2 and 3 — computationally efficient mechanisms, with the tension developed carefully.
Milgrom — Putting Auction Theory to Work	§4 — by a designer of the spectrum auctions, on why the practical choices were made. Library copy.

Module 09 — Matching

Resource	What it covers for this module
Gale & Shapley — College admissions and the stability of marriage (free)	§1 in the original — nine pages, and entirely readable.
Roth — Who Gets What and Why	§3 — the deployments described by the person who built several of them, including the failures. Library copy.
Abdulkadiroglu & Sönmez — School choice: a mechanism design approach (free)	§3's second case — the Boston mechanism's problem, and the reform argument.
Nisan et al., chapter 10 (free PDF)	§§1, 2, and 4 — the computational treatment, including the NP-hard variants.

Module 10 — Voting and Social Choice

Resource	What it covers for this module
Brandt et al. — Handbook of Computational Social Choice (free PDF)	The whole module , free — and the manipulation-complexity chapters cover §4 including the critique.
Arrow — Social Choice and Individual Values	§2 — the original, and the conditions are stated more carefully there than in most summaries. Free in full from the Cowles Foundation, which published it.

Resource	What it covers for this module
Satterthwaite — Strategy-proofness and Arrow's conditions	§3 — and the connection to Arrow is made explicit.
Shoham & Leyton-Brown, chapter 9 (free PDF)	§§1 to 3, free — the rules and the impossibilities, with worked examples.

Module 11 — Fair Division

Resource	What it covers for this module
Brams & Taylor — Fair Division	§§1 and 2 — the procedures, including the moving-knife constructions. Library copy.
Procaccia — Cake cutting algorithms, in the Handbook (free PDF)	§2 — the computational treatment, including the query complexity of envy-freeness.
Caragiannis et al. — The unreasonable fairness of maximum Nash welfare (free)	§3 — a rule that is simultaneously EF1 and efficient, which is a satisfying result.
Spliddit (free)	§§3 and 4 — these algorithms as a working service, which makes the criteria concrete.

Module 12 — Learning in Games

Resource	What it covers for this module
Nisan et al., chapter 4 (free PDF)	§§1 and 2 — learning, regret, and the convergence results, with the proofs.
Roughgarden — Twenty Lectures, lectures 17 to 18 (free)	§2 — no-regret dynamics and the connection to the price of anarchy bounds, which is the key link.
Blum & Mansour — From external to internal regret (free)	§2's swap-regret result , which is what gets you to correlated equilibrium.
Cesa-Bianchi & Lugosi — Prediction, Learning, and Games	§1 — the regret framework in full, and the same material CSCE 669 Module 09 uses. Library copy.

Module 13 — Claiming a Mechanism Works

Resource	What it covers for this module
Roughgarden — Twenty Lectures, the closing lectures (free)	§§1 and 4 — what the guarantees mean and where they stop, stated by somebody careful about it.
Milgrom — Discovering Prices	§4 — a deployed mechanism described by its designer, including what had to be given up and why. Library copy.

Resource	What it covers for this module
Rothkopf — Thirteen reasons why the Vickrey-Clarke-Groves process is not practical	§§1 and 4 — the practical objections to the theoretically optimal mechanism, enumerated.
The ACM EC proceedings (free preprints)	§§1 and 2 — current work, and the better papers state all three of §1's items in the abstract.